

Michael Adu

Embedded Systems Engineer

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A passionate, motivated and project-oriented Embedded Systems Engineer, constantly looking for a challenge and the opportunity to further my knowledge and expand my skills. Skilled in embedded programming languages such as C/C++ and have extensive experience with microcontrollers and SoCs such as Arduinos, Raspberry Pi's, ARM Cortex-M microcontrollers and PIC18 microcontrollers. Looking for opportunities to apply professional skills to various projects through opportunities as an Embedded Systems Engineer.

Experience

2023 – PRESENT

Graduate Embedded Systems Engineer | Skyships Automotive Ltd. | Braintree, United Kingdom

- Designed and developed automotive infotainment and cockpit controllers for high end vehicles, integrating CAN messages with various telltales and warnings using QNX.
- Created several tools to aid in software testing, including a suite of tools to exponentially reduce the duration necessary for calculating and sending automotive CAN messages, performing various critical tests.
- Developed 2D and 3D user interfaces using Kanzi and Lua.

FEBRUARY 2022 – AUGUST 2022

Embedded Engineer and Lead Web Developer | Premium Technologies Ltd. | Accra, Ghana

- Designed and developed a custom Smart Home System using ESP8022 to interface with already established home electrical systems to be implemented in over 3000 homes.
- Created a Linux-based Smart Home Hub running on a Raspberry Pi and a companion phone application using Flutter to enable user control of household appliances and monitoring of CCTV cameras on a local network.

JUNE 2021 – FEBRUARY 2022

Embedded Engineer and Web Developer | Mingo Blox | Accra, Ghana

- Developed a desktop and mobile graphical-based code application for quick and easy programming of Arduino devices for novel engineering students and hobbyists using ReactJS and C/C++, which was marketed to an estimated 30 million users
- Taught the fundamentals of robotics to 30+ young students at the primary level.

Education

2022 - 2023

Masters of Science Embedded Systems Engineering | Graduated with Distinction | Coventry University | Coventry, United Kingdom

2017 - 2021

Bachelors of Science Electrical and Electronic Engineering | Ashesi University | Berekuso, Ghana

Projects

Automated Sign Language Detection Device

- Developing a physical device to recognize sign language hand gestures with the aid of a PIC18F, piezoelectric sensors and machine learning to aid potentially over 430 million audible-impaired individuals with communication.
- Trained a Long Short Term Model Neural Network model to recognize gestures made by the user. The technologies currently in use are C/C++, Proteus, Python and MATLAB.

Yocto GUI

- Developing a graphical user interface for creating Linux Yocto-based operating systems for embedded systems.
- The application aims to bring ease of access to Yocto and provide a centralised workflow when developing operating systems to aid in an industry with an estimated \$116 billion market size. The technologies currently in use are Flutter, NodeJS, and Linux Shell.

Range Sensing Embedded Operating System

- Developed a custom Embedded Linux operating system using Yocto to be deployed on a Raspberry Pi. The operating system comes with a custom device driver for an Ultrasonic Sensor. The technologies used were Yocto, BitBake and C/C++.

Skills

Circuit Programming	Bare Metal C/C++, Assembly, Python, VHDL Debugging
Operating Systems	Linux, RTOS, Yocto, QNX, ROS
Microcontrollers/SoC	PIC18, KL25Z, Arduino, Teles Mote, Raspberry Pi, STM32, ARM Cortex, ESP32
System Testing	Oscilloscopes, Logic Analysers
Web and Application Development	HTML, JavaScript, Flutter/Dart, ReactJS, PHP, SQL, Figma, QT
Communication Protocols	I2C, Serial Peripheral Interface (SPI), UDP, TCP, MQTT, UART, CAN
Project Management	Git, Github
HMI Tools	Kanzi
Others	SolidWorks, Blender, MATLAB, Microsoft Office, Lua

Activities

3D modelling and animation • [Game development](#)

References available upon request